tsvc/vpvtv · vpvtv.c · default · claude-haiku-4-5-20251001 · FASTER

−1 / +15 lines of 175 · 1 parallel region(s) added · 2 of 2 candidate(s) kept · RESTRUCTURING (11 code line(s) added, 1 removed) · restructuring level 0

BEFORE

```
131  #endif
132  }
133
134  static real_t kernel_vpvtv(void)
135  {
136      for (int nl = 0; nl < R; nl++) {
137          for (int i = 0; i < LEN_1D; i++) {
138              a[i] += b[i] * c[i];
139          }
140          pb_mix(nl);
141      }
142      return (real_t)0;
143  }
144
145  int main(int argc, char** argv)
146  {
```

AFTER

```
131  #endif
132  }
133
134  static real_t kernel_vpvtv(void)
135  {
136      real_t* b_fixed = (real_t*)malloc((size_t)LEN_1D * sizeof(real_t));
137      real_t* c_fixed = (real_t*)malloc((size_t)LEN_1D * sizeof(real_t));
138      memcpy(b_fixed, b, (size_t)LEN_1D * sizeof(real_t));
139      memcpy(c_fixed, c, (size_t)LEN_1D * sizeof(real_t));
140
141      for (int nl = 0; nl < R; nl++) {
142          #pragma omp parallel for shared(c_fixed,b_fixed)
143          for (int i = 0; i < LEN_1D; i++) {
144              a[i] += b_fixed[i] * c_fixed[i];
145          }
146          pb_mix(nl);
147          /* Update fixed copies to reflect pb_mix's modifications */
148          long k = ((long)nl * 7919L + 13L) % LEN_1D;
149          b_fixed[k] = b[k];
150          c_fixed[k] = c[k];
151          b_fixed[LEN_1D-1] = b[LEN_1D-1];
152      }
153
154      free(b_fixed);
155      free(c_fixed);
156      return (real_t)0;
157  }
158
159  int main(int argc, char** argv)
160  {
```

file



red = removed green = added yellow = parallel region (#pragma omp) right strip = the whole file: green where changed, yellow where a parallel region now is